

HANLU XU, Ph. D.

Assistant Researcher, Department of Energy and Power Engineering, Tsinghua University, Beijing, China

hanluxu@mail.tsinghua.edu.cn +86 132 1416 9577

Website: <https://hanluxu.github.io/>



■ RESEARCH INTERESTS & EXPERTISE

Research interests and expertise mainly focus on resource valorization and low-carbon energy technologies, including Salt-lake magnesium resource utilization (multiphase flow, heat and mass transfer, and thermochemical coupling); Thermal behavior of furnace in metallurgical processes (multiscale modeling, computational fluid dynamics (CFD), and droplet dynamics); Process intensification and system-level optimization (Aspen Plus process simulation and 4E analysis); and emerging directions in biomass/waste hydrothermal conversion, pyrolysis, gasification, and in situ CO₂ absorption-enhanced hydrogen production. My research aligns with Engineering Science and Technology: thermodynamics and heat transfer, mechanical, energy, and computational engineering.

■ EMPLOYMENT & RESEARCH EXPERIENCE

Assistant Researcher, Department of Energy and Power Engineering, Tsinghua University

(Host: Prof. Hui Zhou)

Jan 2026 – Present

- Developed efficient and low-energy consumption solutions for in-situ CO₂ absorption by biomass enhances hydrogen production; co-supervised two exchange Ph.D. students from Coak University.

Visiting Research Student, School of Energy and Environment, Southeast University

Apr 2024 – Aug 2024

- Designed a visualized droplet evaporation experimental device

Visiting Research Student, School of Mechanical Engineering, Tianjin University

Jul 2023 – Sept 2023

- Completed the International Summer Training Program on Heat Transfer

■ EDUCATION

Ph.D., Energy and Power Engineering, Northeastern University, Shenyang, China

Sept 2021 – Oct 2025

Dissertation: *Evaporation and Decomposition Characteristics of Bischofite Droplets and Applications*,

Supervisor: Prof. Hui Dong Excellent Undergraduate Thesis Award

M.S., Energy and Power Engineering, Northeastern University, Shenyang, China

President's Scholarship Recipient, Graduated one year early.

Sept 2019 – Jun 2021

B.S., Energy and Power Engineering, Shenyang University of Chemical Technology, Shenyang, China

Energy and Power Engineering (Honors) Excellent Undergraduate,

Sept 2015 – Jun 2019

■ SELECTED PUBLICATIONS

(Full list of 20+ publications on Scopus and Website: <https://hanluxu.github.io/>)

- [1] **Hanlu Xu**, Yating Kong, Daokuan Cheng, Liang Zhao, Hui Dong, Menghui Zhang. Optimization of bischofite pyrolysis furnace system based on orthogonal experiments and exergy analysis with assessment of its green economy[J]. *Energy*, 2025, 322: 135773. (First author)
- [2] **Hanlu Xu**, Hui Dong, Liang Zhao, Menghui Zhang, Daokuan Cheng. Isoconversional kinetic analysis of the pyrolysis of Salt Lake industrial waste bischofite with isothermal reaction time predictions[J]. *Process Safety and Environmental Protection*, 2023, 169: 725-735. (First author)
- [3] **Hanlu Xu**, Daokuan Cheng, Liang Zhao, Hui Dong, Bowei Liu. Numerical study of bischofite precursor droplet evaporation and thermal decomposition for spray pyrolysis systems[J]. *Chemical Engineering Science*, 2025, 302: 120680. (First author)
- [4] **Hanlu Xu**, Daokuan Cheng, Liang Zhao, Yingying Yu, Hui Dong. Single bischofite precursor droplet evaporation:

Modeling and simulation covering the effects of decomposition reactions and environmental variables[J]. International Journal of Thermal Sciences, 2025, 208: 109527. (First author)

- [5] **Hanlu Xu**, Daokuan Cheng, Liang Zhao, Hui Dong. Exploring multistep bischofite waste pyrolysis: insights from advanced kinetic analysis and thermogravimetric technique[J]. Environmental Science and Pollution Research, 2024, 31: 13867-13882. (First author)
- [6] **Hanlu Xu**, Hui Dong, Liang Zhao, Daokuan Cheng. Exergy analysis and modeling of pilot-scale pyrolysis for magnesium oxide preparation from salt lake bischofite industrial waste[J]. ACS Omega, 2023, 8(49): 47153-47162. (First author)
- [7] **Hanlu Xu**, Yating Kong, Daokuan Cheng, Yingying Yu, Liang Zhao, Hui Dong, Menghui Zhang. Unraveling the evaporation characteristics of bischofite precursor droplets using suspended droplet evaporation method: experimental studies and modeling[J]. Chemical Engineering Science, 2026, 319: 122316. (First author)
- [8] **Hanlu Xu**, Hui Dong, Liang Zhao, Daokuan Cheng. Study on thermal decomposition behavior and kinetics of industrial waste bischofite[J]. Journal of Northeastern University (Natural Science), 2025, 46(3): 52-59. (First author)
- [9] **Hanlu Xu**, Daokuan Cheng, Liang Zhao, Hui Dong. Optimal operation of pyrolysis in waste bischofite to maximize exergy efficiency and sustainable clean production[C]//6th Euro-Mediterranean Conference for Environmental Integration. Morocco: Springer Publishing, 2024: 94. (First author)
- [10] **Hanlu Xu**, Daokuan Cheng, Liang Zhao, Hui Dong. Numerical simulation of bischofite precursor droplet evaporation considering decomposition reaction[C]//Third Asian Conference on Thermal Sciences. Shanghai: 3rd ACTS Publishing, 2024: 68. (First author)
- [11] Daokuan Cheng, **Hanlu Xu**, Liang Zhao, Hui Dong, Zhijun Zhang. Effect of swirling gas inlet design on particle motion and decomposition in magnesite flash calciner[J]. Chemical Engineering Research and Design, 2024, 206: 386-396. (Co-author)
- [12] Daokuan Cheng, **Hanlu Xu**, Bai Wang, Liang Zhao, Hui Dong, Zhijun Zhang. Kinetic study of magnesite decomposition under different CO₂ pressure conditions and its application in CFD modeling[J]. Chemical Engineering Research and Design, 2025, 217: 440-452. (Co-author)
- [13] Daokuan Cheng, **Hanlu Xu**, Xichen Han, Liang Zhao, Hui Dong, Zhijun Zhang, Mingming Li, Jinhui Zhang. Optimization of operating parameters in a CO₂ self-circulation magnesite flash calciner based on CFD and NSGA-II [J]. Chemical Engineering Research and Design, 2026, 227: 718-728. (Co-author)

■ TALKS & PRESENTATIONS

“Optimal Operation of Pyrolysis in Waste Bischofite to Maximize Exergy Efficiency and Sustainable Clean Production” EMCEI-2024, Cadi Ayyad, Morocco-conference talk (Oral).

“Numerical simulation of bischofite precursor droplet evaporation considering decomposition reaction” ACTS 2024, Shanghai, China-conference talk (Oral).

“In situ CO₂ absorption-enhanced hydrogen production” ACEF 2026, Shenyang, China (Invitation Report)

“Magnesite Decomposition Characteristics in CO₂ Atmosphere” NCMRU 2024, Anshan, China (Invitation Report)

■ HONORS & AWARDS

Outstanding Doctoral Dissertation Award, Northeastern University, 2025.

First prize, 3rd "Dual Carbon" Goal Engineering Thermocatalytic Technology Design Competition, Liaoning Provincial Education Department, 2024.

National Scholarship for Doctoral Students, Ministry of Education of China, 2024.

President's Scholarship, Northeastern University, 2019

National Scholarship for Bachelor Students, Ministry of Education of China, 2018.

■ EDITORIAL & REVIEW EXPERIENCE

Reviewer for Heat and Mass Transfer, Chemical Engineering Science, and Chemical Engineering Journal